Verril Vaz

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Education

REVA University — Bengaluru, India
B.Tech in Computer Science and Engineering – CGPA: 9.33/10 — *2023 – 2027*

- Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Machine Learning
- **Smart India Hackathon Finalist (2025)** – Selected among 50,000+ teams nationwide *(Govt. of India)*

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, Rust, SQL
Frameworks: React.js, Next.js, Node.js, Express.js, FastAPI, Flask, PyTorch
Databases: PostgreSQL, MySQL, MongoDB, SQLite
Tools & Platforms: Git, Docker, REST APIs, Postman, JWT Auth, AWS

Work Experience

Dev Creations & Solutions — Bengaluru, India
Web Developer — *Apr 2025 – Jun 2025*

- Built and tested 8+ REST API endpoints using Node.js and Express.js for production web applications serving real clients
- Implemented JWT-based authentication and session management, securing 100% of API routes against unauthorized access
- Resolved 15+ frontend-backend integration bugs across React components, improving cross-device responsiveness and reducing page load time by **30%**

Projects

Vexa: Donate & Help | *Live on Google Play Store* | *Node.js, Express.js, React Native, Firebase*

- Founded and launched a **production platform** connecting donors with 10+ verified NGOs; engineered the full backend, REST APIs, Firebase real-time database, and end-to-end deployment pipeline
- Serving **80+ registered users** with 100+ downloads; managed live release cycles and scalable backend architecture on Google Play Store

SecurePay – Real-Time Fraud Detection System | *React, TypeScript, FastAPI, XGBoost*

- Built a full-stack fraud detection platform with a React + TypeScript frontend and **FastAPI backend**; integrated an XGBoost ML model for real-time transaction risk scoring with ALLOW / REVIEW / BLOCK decisions
- Engineered multi-signal detection pipeline: velocity analysis, geo-location impossibility checks, spending anomaly detection, and time-based risk scoring
- Designed CORS-enabled REST API with SQLite persistence for transaction history and session management; mobile-responsive UI with Tailwind CSS

verril-learn – ML Unlearning Library | *Python, PyTorch, pip-packaged*

- Built and published on PyPI an open-source ML privacy library enabling **selective data removal from trained AI models** without full retraining – directly addressing data privacy in production AI systems
- Achieved 98.98% accuracy retention on MNIST using Fisher-constrained gradient optimization; supports SimpleCNN (421K params) and ResNet-18 (11.2M params) – production-ready with full documentation

Research & Publications

Machine Unlearning via Fisher-Constrained Weight Manipulation | *Citable Research Artifact (DOI)*

- Designed a surgical unlearning pipeline to remove poisoned/sensitive data from trained neural networks **without full retraining**; applied Fisher Information–guided weight profiling with constrained reverse optimization
- Published as a reproducible research artifact with complete documentation – demonstrates depth in **AI model engineering and data privacy**

Leadership & Activities

Founder & President, Team Sahayak Foundation — My Bharat, Govt. of India
Registered nonprofit affiliated with Ministry of Youth Affairs; community support and grassroots relief

President, Cybersecurity Club — REVA University
Leading technical workshops and campus-wide security initiatives — *2026 – Present*